R-UNIVERSALITY OF THE EVEN SPECIAL ORTHOGONAL ALGEBRA IN TERMS OF THE EVEN CLIFFORD ALGEBRA AND THE DETERMINANT SIGN

DAVE WITTE MORRIS

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Allowed standard prerequisites (editorial).

- The Tits algebras attached to the three non-zero elements of the character group for type D_n, and the identification of the first of them with the even Clifford algebra. Used inside the counted proof at source lines 684-690 and 708-711, attributed by the author to section 27B, type D_n, page 379 of the cited book KnusEtAl-BookOfInvols. Supplies the Brauer classes whose base-change behaviour decides R-universality.
- The division-algebra description of R-universality and the meaning of being an inner form for these orthogonal algebras. The author's lemma labelled RunivIff, invoked in the counted proof at source lines 710-711, together with the standard description of inner forms of type D_n. Turns the computation of the Tits algebras into the two alternatives of the statement.

Source reference. Original lemma 2.1: sources/algebra-fl02-a-cohomological-proof-that-real-re/pinned-original-source.tex, source lines 416–416. Navigation pointer only: the counted proof invokes the author's equivalence for R-universality; that

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lemma has its own card in this lane and its content is declared as a named standard prerequisite.

Source reference. Original definition 3: sources/algebra-fl02-a-cohomological-proof-that-real-re/pinned-original-source.tex, source lines 428–428. Navigation pointer only: the labelled item of that equivalence used by the counted proof.

Source reference. Original lemma 2.3: sources/algebra-fl02-a-cohomological-proof-that-real-re/pinned-original-source.tex, source lines 471–471. Navigation pointer only: the author's lemma on the minimal inner extension; it has its own card in this lane and its content is declared as a named standard prerequisite.

Proposition 0.1. *Let $\mathfrak{g} = \mathfrak{so}_{2k}(B;\mathbb{Q})$, with $k \geq 3$, for some symmetric $B \in \mathrm{GL}_{2k}(\mathbb{Q})$. Then $\mathfrak{g}$ is $\mathbb{R}$-universal if and only if either*

1. *$\mathrm{Cliff}^0_{\mathbb{Q}}(B)$ is split, and $(-1)^k \det B$ is either negative or a square in $\mathbb{Q}$, or*
2. *$\mathrm{Cliff}^0_{\mathbb{Q}}(B)$ does not split over $\mathbb{R}$, and $(-1)^k \det B$ is a square in $\mathbb{Q}$.*

Proof. As in [4, § 27B, type D_n, p. 379], let λ, λ_+, λ_- be dominant weights that represent the three nonzero elements of Z^*, with λ being the highest weight of the natural representation of $\mathfrak{g}$ on $\mathbb{C}^{2k}$. Then $D_{\mathfrak{g},\mathbb{Q}}(\lambda) = \mathbb{Q}$ is trivial [4, p. 379].

Suppose, first, that $\mathfrak{g}$ is an inner form, which means that $(-1)^k \det B$ is a square in $\mathbb{Q}$. Then $\mathrm{Cliff}^0_{\mathbb{Q}}(B)$ is a direct sum of two algebras C^+ and C^- that are Brauer equivalent to the full Clifford algebra [5, Theorem 2.5(3), p. 110] (and are therefore Brauer equivalent to a quaternion algebra). Furthermore, $D_{\mathfrak{g},\mathbb{Q}}(\lambda_\pm)$ is Brauer equivalent to $C^\pm$ [4, p. 379]. Therefore, Lemma 2.1(3) shows that an inner form $\mathfrak{g}$ is:

- automatically $\mathbb{R}$-universal, when $\mathrm{Cliff}^0_{\mathbb{Q}}(B)$ does not split over $\mathbb{R}$, but
- $\mathbb{R}$-universal if and only if $\mathrm{Cliff}^0_{\mathbb{Q}}(B)$ is split, when $\mathrm{Cliff}^0_{\mathbb{Q}}(B)$ splits over $\mathbb{R}$.

Assume, now, that $\mathfrak{g}$ is an outer form. In order for $\mathfrak{g}$ to be $\mathbb{R}$-universal, $\mathfrak{g}$ must remain outer over $\mathbb{R}$ (see Lemma 2.3), which means $(-1)^k \det B < 0$. Let L be the (unique, imaginary) quadratic extension of $\mathbb{Q}$ over which $\mathfrak{g}$ becomes inner. Then $D_{\mathfrak{g},\mathbb{Q}}(\lambda_\pm)$ is Brauer equivalent to $\mathrm{Cliff}^0_{\mathbb{Q}}(B)$ [4, p. 379], which is central simple over L. Since B splits over $\mathbb{R}$ (recall that the quadratic extension L is imaginary), Lemma 2.1(3) implies that $\mathfrak{g}$ is $\mathbb{R}$-universal if and only if $\mathrm{Cliff}^0_{\mathbb{Q}}(B)$ is split. □

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